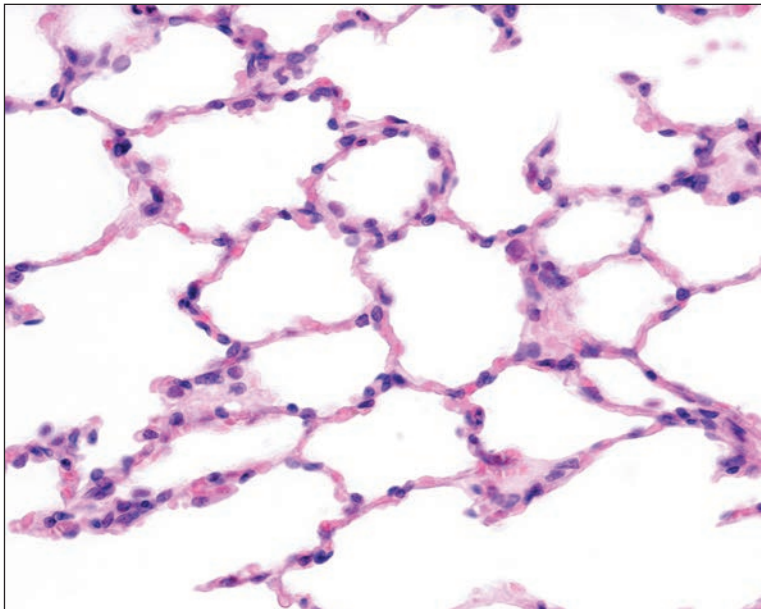


Answer **all** questions.

1. **Image 1.1** shows simple squamous epithelial tissue from the alveoli of the lungs.

Image 1.1



(a) Explain how the shape of the cells is adapted to their function within the tissue. [1]

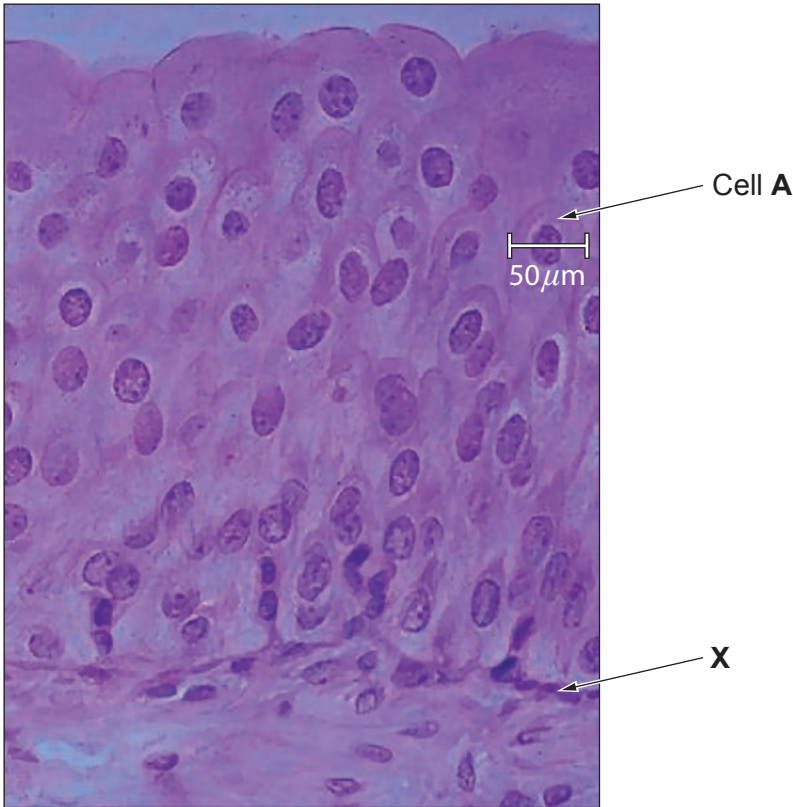
.....

.....



- (b) The urinary bladder is lined with stratified epithelial tissue with a variable number of cell layers. **Image 1.2** shows this tissue when the bladder is relatively empty.

Image 1.2



- (i) Name the structure labelled **X** on **Image 1.2**. [1]
-
- (ii) State why the cells in **Image 1.2** are referred to as a “tissue”. [1]
-
-
- (iii) Explain why some of the cells appear not to have a nucleus in **Image 1.2**. [1]
-
-
- (iv) With reference to **Image 1.1** and **Image 1.2**, describe the main difference between the two types of epithelial tissue. [1]
-
-



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- (c) (i) When the bladder is full, the cells become wider and flatter. Suggest how this property of the epithelial tissue allows the bladder to perform its function. [1]

.....

.....

- (ii) The width of cell **A** in **Image 1.2** is $50\mu\text{m}$. Cell **A** is shown at 42% of its maximum width.
Calculate the maximum width of this cell. **Give your answer to three significant figures.** [2]

Maximum width = μm

- (iii) With reference to the actual width of cell **A**, calculate the magnification of **Image 1.2**. [2]

Magnification = $\times$

10

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